

DocTurn

Requirements Specification & Complete Data Model

Software Requirements + Database Reference

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This document specifies the complete functional and non-functional requirements of the DocTurn platform and documents every table and column in the production data model. Requirements use the convention **FR-** (functional) and **NFR-** (non-functional). The data model is transcribed directly from the Drizzle schema so it stays faithful to the running system.

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1. Introduction

1.1 Purpose & scope

DocTurn (marketed as DoctorHeidi) is a multi-tenant, HIPAA-oriented platform that coordinates patient assignment and secure communication inside hospitals. Emergency-room (ER) physicians admit patients and hand them off to hospitalists; the platform automates routing through a configurable round-robin rotation, delivers real-time notifications across multiple channels, tracks acknowledgement, and maintains a HIPAA-compliant audit trail. It ships as a React/Vite web application and an Expo React Native mobile app that share one Express backend, session model, and PostgreSQL database.

1.2 Actors / roles

Role	Description
director	Hospital administrator — manages providers, schedules, round-robin config, reassignment, and organization settings.
er_director	ER administrator — manages ER physicians and oversees the ER side of assignment flow.
er_doctor / er_physician	Admits patients, creates assignment requests, uses AI-assisted intake, messages providers.
hospitalist	Receives assignments, accepts/declines, manages census and patient cap, communicates with peers.
developer	Platform operator — cross-tenant administration of organizations, users, CMS, integrations, and diagnostics.

1.3 Definitions

Term	Meaning
Tenant / Organization	An independent hospital; all data is isolated by organizationId.
Assignment	A request routing a patient from an ER physician to a hospitalist; has a lifecycle and expiry.
Round-robin	Rotation algorithm that selects the next eligible hospitalist by rotation order, cap, shift, and availability.
PHI	Protected Health Information — patient initials, room, issue summary, messages, and related records.
Census / cap	A hospitalist's current patient count vs. their maximum patient cap.

2. Functional Requirements

2.1 Authentication & sessions

ID	Requirement
FR-1.1	The system shall authenticate users with a username and password using server-side session-based authentication (Passport.js).
FR-1.2	Passwords shall be stored only as salted hashes; plaintext passwords shall never be persisted or logged.
FR-1.3	The system shall support optional two-factor authentication via TOTP authenticator apps and via SMS one-time codes.
FR-1.4	The system shall issue single-use backup codes for account recovery when 2FA is enabled.
FR-1.5	Any configured second factor shall be enforced during the primary login flow, not bypassable through alternate paths.
FR-1.6	The system shall maintain server-side session records, support session expiry/timeout, and allow administrative termination.
FR-1.7	The system shall record last-login time and track active presence/online state per session.

2.2 Multi-tenancy & organizations

ID	Requirement
FR-2.1	Every domain record shall be scoped to an organization; users shall only read or write data within their own organization.
FR-2.2	Organizations shall be identified by a unique code (e.g. MAYO) and be independently configurable (timezone, branding, feature flags).
FR-2.3	Each organization shall configure assignment timeout, round-robin rules, and custom shift-type definitions.
FR-2.4	Per-organization, per-portal feature toggles shall control availability of features such as SMS, AI chatbot, and integrations.
FR-2.5	Developers may operate across tenants for administration; all such access shall be audited.

2.3 User & provider management

ID	Requirement
FR-3.1	Directors shall create, edit, deactivate, and delete provider accounts within their organization.
FR-3.2	The system shall support a self-registration request workflow that requires director/approver review before an account is created.
FR-3.3	A provider-registration approval pipeline shall capture applicant details (name, email, phone, specialty, license, credentials) and track status (pending / approved / rejected / on_hold).
FR-3.4	Role types and their permissions/portal access shall be configurable, including custom roles per organization.

ID	Requirement
FR-3.5	Hospitalist profiles shall capture specialty, phone, patient cap, rotation order, shift type/times, and daily census counters.

2.4 Patient assignment workflow

ID	Requirement
FR-4.1	ER physicians shall create a patient record (initials, room, issue summary) and an assignment request.
FR-4.2	The system shall select the target hospitalist via round-robin, honoring rotation order, patient cap, shift type, on-call and availability filters as configured by the organization.
FR-4.3	ER physicians/directors may also assign a specific hospitalist manually, overriding round-robin.
FR-4.4	Each assignment shall have the lifecycle: pending → accepted / rejected / expired / cancelled, with timestamps for each transition.
FR-4.5	Assignments shall expire automatically after the organization's configured timeout; expiry and rejection shall trigger re-routing to the next eligible provider.
FR-4.6	Directors shall be able to cancel or reassign assignments.
FR-4.7	Accepting an assignment shall update the hospitalist's current patient count and the patient's assigned provider.
FR-4.8	Users shall be able to chat directly on an assignment (assignment-scoped messages with read tracking).

2.5 Notifications

ID	Requirement
FR-5.1	On assignment events, the system shall notify the target provider in real time via WebSocket.
FR-5.2	The system shall send SMS notifications (Twilio) and track delivery and acknowledgement, including the provider message SID.
FR-5.3	The system shall support push notifications (Firebase Cloud Messaging) to registered iOS/Android/web device tokens with priority and deep-link/web-fallback handling.
FR-5.4	Providers shall be able to acknowledge assignments via SMS reply, updating assignment state.
FR-5.5	The system shall support advance/expiry-warning notifications before an assignment times out.

2.6 Messaging & directory

ID	Requirement
FR-6.1	The system shall provide direct, group, and emergency conversations scoped to an organization.
FR-6.2	Messages shall support text and attachments (image/file), replies (threading), editing, and soft deletion.
FR-6.3	Per-recipient delivery status (sent / delivered / read / failed) shall be tracked, with push and SMS-backup flags.
FR-6.4	A provider directory shall show display name, title, specialty, contact, on-call status, and availability.

ID	Requirement
FR-6.5	Directors shall send emergency broadcasts to targeted roles/departments, optionally requiring acknowledgement, with acknowledgement tracking.
FR-6.6	All messaging actions on patient-related content shall be recorded in a message audit log with a reason-for-access where applicable.

2.7 Scheduling

ID	Requirement
FR-7.1	The system shall maintain daily hospitalist schedules (date, shift type, start/end, working flag, notes).
FR-7.2	Round-robin eligibility shall be able to consider schedule/shift data and on-call state.
FR-7.3	The system may integrate with Amion to synchronize provider schedules per organization.

2.8 Hospital resource management

ID	Requirement
FR-8.1	The system shall manage departments with capacity and alert thresholds.
FR-8.2	The system shall track beds (type, status, occupancy) and equipment (type, status, maintenance dates).
FR-8.3	The system shall maintain staff schedules and generate resource alerts (bed capacity, staff shortage, equipment failure) by severity.
FR-8.4	The system shall record resource metrics (e.g. bed occupancy, utilization) over time for monitoring.

2.9 AI assistance

ID	Requirement
FR-9.1	ER physicians and directors shall have access to OpenAI-powered assistance for structured patient-intake extraction from free text.
FR-9.2	The system shall provide an AI monitoring assistant on the developer dashboard for metrics, insights, diagnostics, and alerting.
FR-9.3	AI features shall be gated by per-organization feature flags and shall not expose PHI beyond the requesting tenant.

2.10 Developer dashboard & CMS

ID	Requirement
FR-10.1	Developers shall administer organizations, users, role configuration, portal access, and integrations.
FR-10.2	Developers shall view system logs, diagnostics, and system controls including data-reset operations.
FR-10.3	A Website CMS shall allow editing the marketing landing page (hero, pricing tiers, carousel, testimonials, features, trust badges, SEO/OG metadata, footer) and the contact page, each with a publish toggle.

ID	Requirement
FR-10.4	CMS changes shall record the last modifier and be covered by audit logging.

2.11 Audit & compliance

ID	Requirement
FR-11.1	The system shall maintain a comprehensive audit trail covering authentication, PHI access, assignment operations, user management, and conversation deletion, with risk level.
FR-11.2	PHI access shall be logged separately with accessed fields, purpose, success/denial, and optional duration.
FR-11.3	The system shall record security incidents (failed logins, unauthorized access, suspicious activity) with severity and resolution tracking.

3. Non-Functional Requirements

3.1 Security

ID	Requirement
NFR-1.1	All authorization shall be enforced server-side for every route, WebSocket action, and record, independent of frontend route guards.
NFR-1.2	All reads/writes shall be scoped to the authenticated user's organizationId to prevent cross-tenant access.
NFR-1.3	The server shall set hardened HTTP headers (Helmet) and apply tiered rate limiting on authentication and general API routes.
NFR-1.4	Sessions shall use HttpOnly cookies; traffic shall be served over HTTPS in production.
NFR-1.5	Secrets and integration credentials shall be held server-side and never exposed to clients, logs, or diagnostics.
NFR-1.6	Non-production auth/test helpers shall be unreachable in production.

3.2 Compliance (HIPAA)

ID	Requirement
NFR-2.1	The system shall maintain HIPAA-compliant audit trails for all PHI access and administrative actions.
NFR-2.2	Responses, logs, and admin tools shall never expose password hashes, session identifiers, backup codes, secrets, or PHI to unauthorized users.
NFR-2.3	Message content shall be flagged for encryption and patient-related access shall capture a reason-for-access.

3.3 Performance & availability

ID	Requirement
NFR-3.1	Assignment and messaging events shall propagate to connected clients in real time via WebSocket, with SMS/push fallback.
NFR-3.2	Public and low-auth endpoints shall avoid expensive operations triggerable by unauthenticated users and shall be rate-limited.
NFR-3.3	Production deployment shall support horizontal auto-scaling behind a reverse proxy.

3.4 Usability & accessibility

ID	Requirement
NFR-4.1	The UI shall be responsive and mobile-first, with touch-friendly controls and accessible focus states and contrast support.
NFR-4.2	Copy shall use plain, everyday language; status shall be conveyed with both color and text/icon.

ID	Requirement
NFR-4.3	The web app shall provide PWA capabilities; the mobile app shall maintain feature parity with the web platform.

3.5 Maintainability & data

ID	Requirement
NFR-5.1	Frontend and backend shall share a single typed schema (Drizzle + drizzle-zod) as the source of truth for data shapes.
NFR-5.2	Schema changes shall be applied via the Drizzle push workflow; manual SQL migrations shall not be hand-written.
NFR-5.3	Inputs shall be validated server-side with Zod schemas before persistence.

3.6 External interfaces

Service	Purpose
PostgreSQL	Primary datastore (Drizzle ORM); also session store.
Twilio	Outbound SMS notifications, delivery/ack tracking, SMS 2FA.
OpenAI	AI intake extraction and developer monitoring assistant.
Firebase (FCM)	Push notifications to iOS/Android/web devices.
Amion	Optional per-organization provider schedule integration.

4. Data Model — Conventions

- Source of truth: `shared/schema.ts` (Drizzle ORM). There are **38 tables**: 37 domain tables plus the sessions store.
- **PK** = primary key; **FK→table** = foreign key reference; **NOT NULL** = required. Defaults are shown where defined.
- `serial` ids are auto-incrementing integers; `uuid` ids default to a random UUID.
- `text[]` / `integer[]` are array columns; `jsonb` stores structured JSON.
- Timestamps default to `now()` where indicated; audit/PHI timestamps are server-set.
- Every tenant-scoped table carries `organizationId` (FK→`organizations`) to enforce isolation.

4.1 Table index

#	Table	Domain area
1	<code>sessions</code>	Auth / session store
2	<code>organizations</code>	Tenancy
3	<code>users</code>	Identity & access
4	<code>pending_registrations</code>	Identity & access
5	<code>role_types</code>	Identity & access
6	<code>provider_registrations</code>	Identity & access
7	<code>hospitalists</code>	Providers
8	<code>provider_directory</code>	Providers
9	<code>daily_schedules</code>	Scheduling
10	<code>staff_schedule</code>	Scheduling
11	<code>patients</code>	Clinical
12	<code>assignments</code>	Clinical
13	<code>assignment_messages</code>	Clinical
14	<code>conversations</code>	Messaging
15	<code>messages</code>	Messaging
16	<code>message_delivery_status</code>	Messaging
17	<code>message_threads</code>	Messaging
18	<code>emergency_broadcasts</code>	Messaging
19	<code>broadcast_acknowledgments</code>	Messaging
20	<code>departments / beds / equipment</code>	Resources
21	<code>resource_alerts / resource_metrics</code>	Resources
22	<code>device_tokens / push_notifications</code>	Notifications
23	<code>firebase_configurations</code>	Integrations
24	<code>integration_settings</code>	Integrations

#	Table	Domain area
25	sms_verification_codes	Auth (2FA)
26	audit_logs / phi_access_logs	Compliance
27	security_incidents / message_audit_log	Compliance
28	user_sessions	Compliance / presence
29	system_settings / portal_configurations	Configuration
30	landing_page_settings / contact_page_settings	CMS

5. Tenancy & Identity

Sessions (sessions)

Express session store (connect-pg-simple). Required for server-side sessions.

Column	Type	Constraints & notes
sid	varchar	PK. Session id.
sess	jsonb	NOT NULL. Serialized session payload.
expire	timestamp	NOT NULL. Expiry; indexed (IDX_session_expire).

Organizations (organizations)

Top-level tenant. All domain data is isolated by this record.

Column	Type	Constraints & notes
id	serial	PK
name	text	NOT NULL
code	text	NOT NULL, unique. Tenant code (e.g. MAYO).
address	text	Optional.
phoneNumber	text	Optional.
amionFacilityId	text	Amion integration facility id.
amionApiKey	text	Org-specific Amion API key.
timezone	text	NOT NULL, default America/New_York.
assignmentTimeoutMinutes	integer	NOT NULL, default 10. Assignment expiry.
roundRobinOnCallOnly	boolean	NOT NULL, default false.
roundRobinActiveOnly	boolean	NOT NULL, default false.
roundRobinShiftTypes	text[]	Default [rounding, swing, nocturnist].
shiftTypeDefinitions	jsonb	Custom shift-type definitions (id, name, times, color).
isActive	boolean	NOT NULL, default true.
createdAt	timestamp	Default now().
updatedAt	timestamp	Default now().

Users (users)

All identities across roles. Includes optional Replit-auth and 2FA fields.

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
username	text	NOT NULL
password	text	NOT NULL. Salted hash.
role	text	NOT NULL. director / er_doctor / hospitalist / developer / er_director.
name	text	NOT NULL
phoneNumber	text	Optional.
specialty	text	For hospitalists.
isActive	boolean	NOT NULL, default true.
replitUserId	varchar	Unique. Optional Replit id.
email	varchar	Unique. Optional.
firstName	varchar	Optional.
lastName	varchar	Optional.
profileImageUrl	varchar	Optional.
twoFactorEnabled	boolean	NOT NULL, default false.
twoFactorSecret	text	TOTP secret.
smsVerificationEnabled	boolean	NOT NULL, default false.
backupCodes	text[]	2FA recovery codes.
lastLoginAt	timestamp	Optional.
createdAt	timestamp	Default now().
updatedAt	timestamp	Default now().

Pending registrations (`pending_registrations`)

Self-service signup requests awaiting director approval.

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
username	text	NOT NULL
password	text	NOT NULL. Hashed.
name	text	NOT NULL
requestedAt	timestamp	Default now().
status	text	NOT NULL, default pending. pending / approved / rejected.
reviewedBy	integer	FK→users.
reviewedAt	timestamp	Optional.
rejectionReason	text	Optional.

Role types (`role_types`)

Configurable RBAC role definitions, including custom per-org roles.

Column	Type	Constraints & notes
id	serial	PK
name	text	NOT NULL
description	text	NOT NULL, default "".
displayName	text	NOT NULL
accessLevel	text	NOT NULL. director / staff / custom.
permissions	text	NOT NULL, default '[]'. JSON array.
portalAccess	text	NOT NULL, default '[]'. JSON array.
organizationId	integer	FK→organizations (null = system).
isSystemRole	boolean	Default false.
isActive	boolean	Default true.
createdAt	timestamp	Default now().
updatedAt	timestamp	Default now().

Provider registrations (provider_registrations)

Credentialed provider onboarding pipeline with approver review.

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
requestedRole	text	NOT NULL
applicantName	text	NOT NULL
applicantEmail	text	NOT NULL
applicantPhone	text	Optional.
specialty	text	Optional.
licenseNumber	text	Optional.
credentials	text	JSON (education, certifications).
requestedBy	integer	FK→users.
approverRole	text	NOT NULL
status	text	NOT NULL, default pending. pending / approved / rejected / on_hold.
reviewedBy	integer	FK→users.
reviewNotes	text	Optional.
approvedAt	timestamp	Optional.
rejectedAt	timestamp	Optional.
createdUserId	integer	FK→users. Set after approval.
createdAt	timestamp	Default now().
updatedAt	timestamp	Default now().

6. Providers & Scheduling

Hospitalists (hospitalists)

Hospitalist provider profile with census, cap, and shift data.

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
userId	integer	NOT NULL, FK→users.
specialty	text	NOT NULL
phoneNumber	text	NOT NULL
isWorkingToday	boolean	NOT NULL, default false.
currentPatientCount	integer	NOT NULL, default 0.
patientCap	integer	NOT NULL, default 15.
rotationOrder	integer	NOT NULL, default 0. Round-robin order.
shiftType	text(enum)	NOT NULL, default rounding. rounding / swing / nocturnist.
shiftStartTime	text	NOT NULL, default 07:00.
shiftEndTime	text	NOT NULL, default 19:00.
admissionsToday	integer	NOT NULL, default 0.
dischargesPlanned	integer	NOT NULL, default 0.
amionPhysicianId	text	Amion physician identifier.

Provider directory (provider_directory)

Searchable directory entry for messaging and contact.

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
userId	integer	NOT NULL, FK→users.
displayName	text	NOT NULL
title	text	NOT NULL. Dr. / RN / PA, etc.
specialty	text	Optional.
department	text	Optional.
phoneNumber	text	Optional.
pagerNumber	text	Optional.
email	text	Optional.
isOnCall	boolean	Default false.
availability	text	Default available. available / busy / do_not_disturb / offline.
lastSeen	timestamp	Optional.
profileImageUrl	text	Optional.
bio	text	Optional.
languages	text[]	Languages spoken.
certifications	text[]	Certifications.
isVisible	boolean	Default true.
messagePreferences	text	Default '{}'. JSON notification prefs.
createdAt	timestamp	Default now().
updatedAt	timestamp	Default now().

Daily schedules (daily_schedules)

Per-hospitalist daily shift schedule.

Column	Type	Constraints & notes
id	serial	PK
hospitalistId	integer	NOT NULL, FK→hospitalists.
scheduleDate	text	NOT NULL. YYYY-MM-DD.
shiftType	text(enum)	NOT NULL. rounding / swing / nocturnist / off.
shiftStartTime	text	NOT NULL
shiftEndTime	text	NOT NULL
isWorking	boolean	NOT NULL, default true.
notes	text	Optional.
createdAt	timestamp	NOT NULL, default now().
updatedAt	timestamp	NOT NULL, default now().

Staff schedule (staff_schedule)

General staff shift schedule by department.

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
userId	integer	NOT NULL, FK→users.
departmentId	integer	FK→departments.
shiftStart	timestamp	NOT NULL
shiftEnd	timestamp	NOT NULL
shiftType	text	NOT NULL. day / night / swing.
status	text	NOT NULL, default scheduled. scheduled / active / completed / cancelled.
createdAt	timestamp	Default now().
updatedAt	timestamp	Default now().

7. Clinical Core

Patients (patients)

Patient admitted from the ER. Carries PHI (initials, room, issue).

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
initials	varchar(10)	NOT NULL. PHI.
roomNumber	text	NOT NULL. PHI.
issueSummary	text	NOT NULL. PHI.
assignedHospitalistId	integer	FK→hospitalists.
erDoctorId	integer	NOT NULL, FK→users.
assignedAt	timestamp	NOT NULL, default now().
status	text	NOT NULL, default active. active / completed / transferred.

Assignments (assignments)

Routing of a patient to a hospitalist with lifecycle and SMS tracking.

Column	Type	Constraints & notes
id	serial	PK
patientId	integer	NOT NULL, FK→patients.
hospitalistId	integer	NOT NULL, FK→users.
erDoctorId	integer	NOT NULL, FK→users.
assignedAt	timestamp	NOT NULL, default now().
notificationSent	boolean	NOT NULL, default false.
advanceNotificationSent	boolean	NOT NULL, default false.
status	text(enum)	NOT NULL, default pending. pending / accepted / rejected / expired / cancelled.
hospitalistResponse	text	Optional response.
respondedAt	timestamp	Optional.
reassignedAt	timestamp	Optional.
expiresAt	timestamp	Expiry time.
smsDelivered	boolean	NOT NULL, default false.
smsDeliveredAt	timestamp	Optional.
smsAcknowledged	boolean	NOT NULL, default false.
smsAcknowledgedAt	timestamp	Optional.
smsSid	text	Twilio message SID.

Assignment messages (assignment_messages)

Chat scoped to a single assignment.

Column	Type	Constraints & notes
id	serial	PK
assignmentId	integer	NOT NULL, FK→assignments.
senderId	integer	NOT NULL, FK→users.
content	text	NOT NULL
messageType	text	NOT NULL, default assignment_chat.
createdAt	timestamp	NOT NULL, default now().
readAt	timestamp	Optional.
isRead	boolean	NOT NULL, default false.

8. Messaging & Broadcasts

Conversations (conversations)

Direct / group / emergency conversation container.

Column	Type	Constraints & notes
id	uuid	PK, default random.
organizationId	integer	NOT NULL, FK→organizations.
type	text	NOT NULL. direct / group / emergency.
subject	text	Optional.
participants	integer[]	NOT NULL. User ids.
isArchived	boolean	Default false.
isEncrypted	boolean	Default true.
priority	text	Default normal. low / normal / high / urgent.
departmentId	integer	FK→departments.
patientId	integer	FK→patients.
createdBy	integer	NOT NULL, FK→users.
createdAt	timestamp	Default now().
updatedAt	timestamp	Default now().

Messages (messages)

Individual message; supports attachments, replies, edit, soft-delete.

Column	Type	Constraints & notes
id	uuid	PK, default random.
conversationId	uuid	NOT NULL, FK→conversations.
senderId	integer	NOT NULL, FK→users.
content	text	NOT NULL. Encrypted content.
messageType	text	Default text. text / image / file / location / contact.
attachmentUrl	text	Optional.
attachmentType	text	MIME type.
isEncrypted	boolean	Default true.
encryptionKey	text	Message-specific key.
priority	text	Default normal.
isEdited	boolean	Default false.
editedAt	timestamp	Optional.
parentMessageId	uuid	FK→messages (reply).
isDeleted	boolean	Default false.
deletedAt	timestamp	Optional.
sentAt	timestamp	Default now().
createdAt	timestamp	Default now().

Message delivery status (message_delivery_status)

Per-recipient delivery/read state. Table-level constraint: **UNIQUE (messageId, userId)** — one status row per recipient per message.

Column	Type	Constraints & notes
id	serial	PK
messageId	uuid	NOT NULL, FK→messages.
userId	integer	NOT NULL, FK→users.
status	text	NOT NULL. sent / delivered / read / failed.
deviceType	text	mobile / web / sms.
deliveredAt	timestamp	Optional.
readAt	timestamp	Optional.
failureReason	text	Optional.
pushNotificationSent	boolean	Default false.
smsBackupSent	boolean	Default false.
createdAt	timestamp	Default now().

Message threads (message_threads)

Lightweight thread index within a conversation.

Column	Type	Constraints & notes
id	uuid	PK, default random.
conversationId	uuid	NOT NULL, FK→conversations.
subject	text	Optional.
participantIds	integer[]	NOT NULL
lastMessageId	uuid	FK→messages.
lastActivity	timestamp	Default now().
isActive	boolean	Default true.
createdAt	timestamp	Default now().

Emergency broadcasts (emergency_broadcasts)

Org-wide or targeted urgent broadcast.

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
senderId	integer	NOT NULL, FK→users.
title	text	NOT NULL
message	text	NOT NULL
severity	text	NOT NULL. info / warning / critical / emergency.
targetAudience	text[]	NOT NULL. Roles or user ids.
departments	integer[]	Department ids.
requiresAcknowledgment	boolean	Default false.
expiresAt	timestamp	Optional.
isActive	boolean	Default true.
sentAt	timestamp	Default now().
createdAt	timestamp	Default now().

Broadcast acknowledgments (broadcast_acknowledgments)

Per-user acknowledgement of a broadcast.

Column	Type	Constraints & notes
id	serial	PK
broadcastId	integer	NOT NULL, FK→emergency_broadcasts.
userId	integer	NOT NULL, FK→users.
acknowledgedAt	timestamp	Default now().
response	text	Optional.
location	text	Optional.

9. Hospital Resources

Departments (departments)

Hospital department with capacity and alert threshold.

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
name	text	NOT NULL
code	text	NOT NULL. ER / ICU / MED / SURG.
floor	text	Optional.
capacity	integer	NOT NULL, default 0.
alertThreshold	integer	NOT NULL, default 80 (%).
isActive	boolean	NOT NULL, default true.
createdAt	timestamp	Default now().
updatedAt	timestamp	Default now().

Beds (beds)

Individual bed and its occupancy state.

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
departmentId	integer	NOT NULL, FK→departments.
roomNumber	text	NOT NULL
bedNumber	text	NOT NULL
bedType	text	NOT NULL. regular / icu / telemetry / isolation.
status	text	NOT NULL, default available. available / occupied / cleaning / maintenance.
patientId	integer	FK→patients.
assignedAt	timestamp	Optional.
isActive	boolean	NOT NULL, default true.
createdAt	timestamp	Default now().
updatedAt	timestamp	Default now().

Equipment (equipment)

Tracked medical equipment and maintenance.

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
departmentId	integer	FK→departments.
name	text	NOT NULL
type	text	NOT NULL. ventilator / monitor / pump / defibrillator.
serialNumber	text	Optional.
status	text	NOT NULL, default available. available / in_use / maintenance / out_of_order.
location	text	Optional.
lastMaintenanceDate	timestamp	Optional.
nextMaintenanceDate	timestamp	Optional.
isActive	boolean	NOT NULL, default true.
createdAt	timestamp	Default now().
updatedAt	timestamp	Default now().

Resource alerts (resource_alerts)

Alerts raised on resource thresholds/failures.

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
alertType	text	NOT NULL. bed_capacity / staff_shortage / equipment_failure.
severity	text	NOT NULL. low / medium / high / critical.
title	text	NOT NULL
message	text	NOT NULL
resourceType	text	NOT NULL. bed / staff / equipment.
resourceId	integer	Reference to the resource.
departmentId	integer	FK→departments.
isResolved	boolean	NOT NULL, default false.
resolvedAt	timestamp	Optional.
resolvedBy	integer	FK→users.
createdAt	timestamp	Default now().
updatedAt	timestamp	Default now().

Resource metrics (resource_metrics)

Time-series resource utilization samples.

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
departmentId	integer	FK→departments.
metricType	text	NOT NULL. bed_occupancy / staff_count / equipment_utilization.
metricValue	integer	NOT NULL
maxValue	integer	NOT NULL
timestamp	timestamp	NOT NULL, default now().
createdAt	timestamp	Default now().

10. Notifications & Integrations

Device tokens (device_tokens)

Registered push tokens per user/device.

Column	Type	Constraints & notes
id	serial	PK
userId	integer	NOT NULL, FK→users.
organizationId	integer	NOT NULL, FK→organizations.
token	text	NOT NULL
deviceType	text	NOT NULL. ios / android / web.
deviceInfo	text	JSON device details.
isActive	boolean	Default true.
lastUsed	timestamp	Default now().
createdAt	timestamp	Default now().

Push notifications (push_notifications)

Outbound push notification records and status.

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
userId	integer	NOT NULL, FK→users.
title	text	NOT NULL
body	text	NOT NULL
data	text	JSON payload.
deepLink	text	In-app link.
webFallbackUrl	text	Browser fallback.
notificationType	text	NOT NULL. assignment / urgent / reminder / system.
priority	text	Default normal.
status	text	Default pending. pending / sent / delivered / failed.
sentAt	timestamp	Optional.
deliveredAt	timestamp	Optional.
errorMessage	text	Optional.
createdAt	timestamp	Default now().

Firestore configurations (firebase_configurations)

Per-org Firebase project credentials for FCM.

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
projectId	text	NOT NULL
apiKey	text	NOT NULL
appId	text	NOT NULL
messagingSenderId	text	Optional.
serverKey	text	Server-side push key.
vapidKey	text	Web push key.
isActive	boolean	Default true.
lastTestDate	timestamp	Optional.
createdAt	timestamp	Default now().
updatedAt	timestamp	Default now().

Integration settings (integration_settings)

Generic per-org integration config (firebase / twilio / epic / amion).

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
integrationType	text	NOT NULL
isEnabled	boolean	Default false.
configuration	text	NOT NULL, default '{}'. JSON.
lastSync	timestamp	Optional.
syncStatus	text	Default inactive. active / inactive / error.
errorLog	text	Optional.
createdAt	timestamp	Default now().
updatedAt	timestamp	Default now().

SMS verification codes (`sms_verification_codes`)

One-time codes for SMS 2FA / setup / reset.

Column	Type	Constraints & notes
id	serial	PK
userId	integer	NOT NULL, FK→users.
phoneNumber	text	NOT NULL
code	text	NOT NULL
purpose	text	NOT NULL. login / setup / reset.
isUsed	boolean	Default false.
expiresAt	timestamp	NOT NULL
createdAt	timestamp	Default now().

11. Compliance, Sessions & Configuration

Audit logs (audit_logs)

Primary HIPAA audit trail for actions across the system.

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
userId	integer	FK→users.
action	text	NOT NULL. login / view_patient / create_user / access_phi ...
resourceType	text	user / patient / assignment / organization.
resourceId	text	Affected resource id.
ipAddress	text	Optional.
userAgent	text	Optional.
sessionId	text	Optional.
details	text	JSON context.
timestamp	timestamp	NOT NULL, default now().
riskLevel	text(enum)	Default low. low / medium / high / critical.
geolocation	text	Optional.

PHI access logs (phi_access_logs)

Dedicated log of PHI access events.

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
userId	integer	NOT NULL, FK→users.
patientId	integer	FK→patients.
accessType	text	NOT NULL. view / edit / create / delete / export.
dataFields	text[]	Fields accessed.
purpose	text	Medical purpose.
timestamp	timestamp	NOT NULL, default now().
duration	integer	Seconds.
ipAddress	text	Optional.
successful	boolean	NOT NULL, default true.
denialReason	text	If denied.

Security incidents (security_incidents)

Detected security events with resolution tracking.

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
incidentType	text	NOT NULL. failed_login / unauthorized_access / data_breach / suspicious_activity.
severity	text(enum)	NOT NULL. low / medium / high / critical.
userId	integer	FK→users.
ipAddress	text	Optional.
description	text	NOT NULL
status	text(enum)	NOT NULL, default open. open / investigating / resolved / false_positive.
reportedAt	timestamp	NOT NULL, default now().
resolvedAt	timestamp	Optional.
assignedTo	integer	FK→users.
resolution	text	Optional.
affectedData	text	Optional.

Message audit log (message_audit_log)

HIPAA audit trail specific to messaging.

Column	Type	Constraints & notes
id	serial	PK
organizationId	integer	NOT NULL, FK→organizations.
messageId	uuid	FK→messages.
conversationId	uuid	FK→conversations.
userId	integer	NOT NULL, FK→users.
action	text	NOT NULL. create / read / edit / delete / export / share.
details	text	JSON.
ipAddress	text	Optional.
userAgent	text	Optional.
deviceInfo	text	Optional.
patientId	integer	FK→patients.
reasonForAccess	text	Required for patient-related messages.
timestamp	timestamp	Default now().

User sessions (user_sessions)

Application-level session + presence tracking (distinct from the session store).

Column	Type	Constraints & notes
id	serial	PK
userId	integer	NOT NULL, FK→users.
sessionToken	text	NOT NULL, unique.
ipAddress	text	NOT NULL
userAgent	text	Optional.
isActive	boolean	NOT NULL, default true.
lastActivity	timestamp	NOT NULL, default now().
createdAt	timestamp	NOT NULL, default now().
expiresAt	timestamp	NOT NULL
terminatedAt	timestamp	Optional.
terminationReason	text	timeout / logout / admin_termination / security_breach.
isOnline	boolean	NOT NULL, default false.
lastSeen	timestamp	Default now().
activeConnectionCount	integer	NOT NULL, default 0.

System settings (system_settings)

Global key/value settings.

Column	Type	Constraints & notes
id	serial	PK
key	text	NOT NULL, unique.
value	text	NOT NULL
updatedAt	timestamp	NOT NULL, default now().

Portal configurations (portal_configurations)

Per-org, per-portal feature toggles.

Column	Type	Constraints & notes
id	serial	PK
portalId	text	NOT NULL. developer / director / er_director ...
organizationId	integer	NOT NULL, FK→organizations.
featureId	text	NOT NULL. sms_notifications / ai_chatbot ...
enabled	boolean	NOT NULL, default true.
config	text	JSON feature config.
createdAt	timestamp	NOT NULL, default now().
updatedAt	timestamp	NOT NULL, default now().

12. Content Management (CMS)

Landing page settings (landing_page_settings)

Editable marketing landing-page content (singleton-style row). Every text column has a seeded default value in the schema.

Column	Type	Constraints & notes
id	serial	PK
heroTitle	text	NOT NULL. Default 'Transform Patient Assignments'.
heroSubtitle	text	NOT NULL. Default seeded subtitle copy.
heroCta1Text	text	NOT NULL. Default 'Start Free Trial'.
heroCta1Link	text	NOT NULL. Default '#pricing'.
heroCta2Text	text	NOT NULL. Default 'Watch Demo'.
heroCta2Link	text	NOT NULL. Default '#demo'.
heroCheckmarks	jsonb	NOT NULL. Default bullet list.
seoTitle	text	NOT NULL. Default SEO title.
seoDescription	text	NOT NULL. Default SEO description.
seoKeywords	text	NOT NULL. Default SEO keywords.
ogTitle	text	NOT NULL. Default Open Graph title.
ogDescription	text	NOT NULL. Default Open Graph description.
ogImage	text	Default '/og-image.png'.
pricingTiers	jsonb	NOT NULL. Default array of tiers (name, price, features...).
carouselSlides	jsonb	NOT NULL. Default array of slides.
testimonials	jsonb	NOT NULL. Default array of testimonials.
featuresTitle	text	NOT NULL. Default features heading.
featuresSubtitle	text	NOT NULL. Default features subheading.
trustBadges	jsonb	NOT NULL. Default logos/labels.
footerCopyright	text	NOT NULL. Default 'DocTurn by DoctorHeidi'.
isPublished	boolean	NOT NULL, default true.
lastModifiedBy	integer	FK→users.
createdAt	timestamp	Default now().
updatedAt	timestamp	Default now().

Contact page settings (`contact_page_settings`)

Editable contact-page content and form copy. Every text column has a seeded default value in the schema.

Column	Type	Constraints & notes
id	serial	PK
sectionTitle	text	NOT NULL. Default 'Get in Touch'.
sectionSubtitle	text	NOT NULL. Default seeded subtitle.
email	text	NOT NULL. Default 'hello@doctorheidi.com'.
phone	text	NOT NULL. Default '+1 (555) 123-4567'.
office	text	NOT NULL. Default 'San Francisco, CA'.
responseTimeMessage	text	NOT NULL. Default response-time copy.
formNamePlaceholder	text	NOT NULL. Default 'Dr. Jane Smith'.
formEmailPlaceholder	text	NOT NULL. Default 'jane@hospital.com'.
formOrgPlaceholder	text	NOT NULL. Default 'Metro General Hospital'.
formMessagePlaceholder	text	NOT NULL. Default 'Tell us about your needs...'
formSubmitText	text	NOT NULL. Default 'Send Message'.
formThankYouMessage	text	NOT NULL. Default thank-you copy.
isPublished	boolean	NOT NULL, default true.
lastModifiedBy	integer	FK→users.
createdAt	timestamp	Default now().
updatedAt	timestamp	Default now().

12.1 Key relationships

- **organizations** is the tenant root; nearly every table references it via `organizationId`.
- **users** 1→N **hospitalists**, **patients** (as ER doctor), and **assignments**.
- **patients** 1→N **assignments**; an assignment references the patient, the hospitalist, and the ER doctor.
- **conversations** 1→N **messages** 1→N **message_delivery_status** (unique per recipient); messages self-reference via `parentMessageId` for replies.
- **departments** 1→N **beds**, **equipment**, **resource_alerts**, and **resource_metrics**.
- **emergency_broadcasts** 1→N **broadcast_acknowledgments**.